L2.3 Solutions to a system of linear equations with an invertible coefficient matrix

Transcript Language

English

Hello and welcome to the maths 2 component of the online BSC degree in Data Science. In the

last video we have seen something called Cramer's rule which was used to solve a system of equations

in which the coefficient matrix is square and its determinant is non-zero.

Today we are going to continue on that theme and we are go[illegible]o talk about the solution of a system of linear equations with an invertible coefficient matrix. So, we hinted on this

in the last video and we want to continue with that theme. So, for quite some time we will talk

first about the invertibility of a square matrix, what is its relation with the determinant and then

move on to the answer to this question about what is the solution and how we can get it.

So, let us recall quickly what is a square matrix? A square matrix is a matrix in

which the number of rows is the same as the number of columns. So, here is an example,

so the first one is a 3 by 3 example, the second one is a 2 by 2 example. So, a non

example would be something which has 2 rows and 3 columns or let us say 5 rows and 8 columns.

So, let us recall what is the inverse of a square matrix. So, if you have an n by n matrix then the

inverse of A is another n by n matrix B such that A times B is B times A is identity. Now, I will

point out that B has to be n by n otherwise this multiplication on both sides is not defined as we, the number of columns of the first matrix has to be the same as number

of rows of the second matrix in order for them to be multiplied. So, that forces B

to have the same number of rows and columns as A, if this identity were to be satisfied

and recall of course, that the identity matrix is the diagonal matrix with all ones.

So, the inverse is denoted by A inverse, now I am using already in this The inverse of a square

matrix, so I should qualify the The before that, let us see an example, so here is an example,

so you have the matrix 4, 7, 2 ,6 2 by 2 matrix and you can take B to be 0.6, minus 0.7,

minus 0.2, 0.4 of course, you can wonder where did I come with these values from, was it trial

and error or how did I come up with them. So, that we will study that in a few slides.

Anyhow if you multiply these you get identity and if you multiply it on the other side also

you get identity, so this second matrix is the inverse of the first matrix and similarly, the first matrix is the inverse of the second matrix. So, if I call this A and this B

then A is B inverse and B is A inverse. Now, how do I know that

there is no other matrix which satisfies the same thing? So, maybe A inverse B has this and then

some other C also could have the same property. So, how do I know that A inverse is this matrix

and not some other matrix C? So, this is what is called uniqueness, so the inverse is unique

which is why I am using the word The and this is a standard argument in something called group theory

but let us go through it now. So, suppose there are two

matrices with this property so AB is BA is the identity and AC is CA is the identity,

then what do we do? We compute this matrix C times A times B, first of all note that

all of them are n by n matrices, so this product makes sense, second one of the properties that we

saw for matrix multiplication was that it does not matter which order you, it was that, it does not

matter whether you first do A times B or first do C times A and then multiply that to B.

So, in other words we could either do this or we could do this and they are both the same,

but let us compute each of these and see what we get. so, if you first do A times B, then you

get well A times B is the identity matrix, so C times the identity matrix but anything times

identity is itself. So, C times identity is C, on the other hand if you first compute C times A

you get the identity matrix times B, but identity times any matrix is itself, so we get B.

So, what did this show? This showed that B must be C. So, in other words if two matrices satisfy this

relation that A times B is B times A is identity, then they must be the same, so we are saying that

the inverse is unique. Of course, we do not know if it exists or not, as we saw

there was a condition for it to exist. Namely, we saw that the determinant must

be non-zero, how did that come about? That was because we saw that the determinant is

multiplicative, meaning determinant of A times B is determinant of A times determinant of B.

So, if you take determinant of A times A inverse that is determinant of the identity which is 1,

so that means determinant of A inverse times determinant of A is 1,

and that means in particular that neither of them can be

0, they have to be non-zero otherwise the product cannot be 1, 0 times anything is 0 as we know.

So, this is a necessary condition for a matrix to be invertible. So, if the determinant

is 0, so determinant of A is 0, then the matrix A cannot be invertible, that means it cannot have an

inverse. So, we say a matrix is invertible if it has an inverse. So, the conclusion here is

that The inverse of A exists implies that the determinant of A has to be non-zero,

we can go the other way and ask what about the converse? So, if the determinant is non-zero

is A invertible, meaning does an inverse of A exist? So let us maybe do an example.

So, let us do a 2 by 2 example. So, suppose A is a, b, c, d and I am trying to look for

its inverse and what do I know? I know determinant of A is non-zero, so determinant of A is well let us calculate what that is, that is a d minus b c,

that is how we define the determinant of a 2 by 2 matrix. So, now I know that this is non-zero,

that is what I am given. So, now let us try and find an inverse. What does

the inverse satisfy? So, the inverse satisfies that if you take a, b, c, d and if I multiply

suppose, there is an inverse and it is called e, f, g, h

then I should get 1, 0, 0, 1. So, that means I will get a bunch of equations and I can try

to solve them. So, what equations do I get? I get ae plus bg is 1,

af plus bh is 0 and then ce plus dg is 1 and cf plus dh is 0, my bad, ce plus dg is

0 and not 1 and cf plus dh is 1. So, this is what we get, these four equations are what

we get by multiplying the left hand side, that is how we do product of matrices.

Maybe I will multiply the first equation by d. So, ade plus bdg is d

and maybe I will multiply the third equation by b, so bce plus bdg

is 0, that is what I get. So, now I can subtract out the bdg, so if I do that what do I get?

I get ad minus bc times e is d, so what does that tell me? That tells me that e is

d divided by ad minus bc and this is the important part, I can divide because ad minus

bc is non-zero, that is the assumption otherwise I would not have been able to solve this at all.

And then you can work out the rest, so from here what we will get is that the inverse of A

has this form, so it is 1 by ad minus bc times the matrix da and I hope my signs are correct, so

I have c here and a b here and then my bad, this should be c here with a minus sign and

a minus b over here. So, my claim is A inverse is going to be of this form and I will leave you to

check that the other entries give you exactly this. So, you can see that for a 2 by 2 matrix

if the determinant is non-zero, then indeed the matrix is invertible, it has an inverse,

so can I go beyond? So, the converse is true, at least for 2 by 2 matrices,

so what we are going to see next is how to go beyond the 2 by 2 case.

So, I am going to introduce a few notations, so the adjugate of a square matrix,

so what is that? So, remember in the previous video, we defined something called minors and

cofactors, so the i, j-th minor is the determinant of the sub matrix obtained by deleting the i-th

row and the j-th column and we denoted it by an Mij. And the i, j-th cofactor was

defined as multiplying the minor, the i, j-th minor by minus 1 to the power i plus j.

So, we can create a matrix called the cofactor matrix, so the cofactor matrix has i, j-th

entry Cij and then we define the adjugate of this matrix A that you started with as the transpose of

this cofactor matrix. So, remember that we defined transpose, so transpose was when you take a matrix

and you flip the entries about its diagonals, you reflect the entries about the diagonal.

So, let us see an example of the computation of the adjugate matrix. So, here is the matrix

A which is 1, 2, 3 in the first row 0, 2, 8 in the second row, 5, 6, 0 in the third row. Let us compute its minors first, well let us compute its determinant first, determinant of A is

well 1 times 2 times 0 minus 8 times 6 minus two times 0 times 0 minus 8 times 5

plus 3 times 0 times 6 minus 2 times 5 and if you work that out that is minus

48 plus 80 minus 30, so that is 2. So, let us now work out the minors, so

well the minors are written here, I guess you have to work out these minors by yourself. So,

just to check that the M11 is correct, that is exactly the number appearing

in the bracket corresponding to 1 in that expression for the determinant of A.

so, that is minus 48, 1 times minus 48, the first term in that expression for determinant of

that is minus 1, that is exactly M11, why is that M11? Because that is what you get by

hiding the first row and the first column, so you get 2, 8, 6, 0, 2 times 0 minus 8 times 6.

So, you can check the other entries, so the other minors and hopefully you will get these numbers.

So, now let us write down the cofactor matrix, so the cofactor matrix remember is given by

taking these minors and putting them into the corresponding positions but

by multiplying each of them by a minus 1 to the power i plus j. so,

the first term remains the same because it is minus 1 to the power 1 plus 1, so it is minus 48,

the second term comes with a minus sign so its minus of minus 40 which is 40, third term is again

1 times minus 10 so just minus 10 and so on. So, you have 18, minus 15, 4 and then 10, minus 8,

2. So, this is your cofactor matrix, again this is something you have to compute by yourself

but I hope the ideas are clear, that to compute the adjugate you have to compute the

minors first, to compute the minors for Mij you hide the i, j-th row in the j-th column

and compute the resulting matrix and then put those into multiply that by minus 1 to

the power i plus j and put that into a matrix that will give you the cofactor matrix.

And then to get the adjugate matrix you take this same cofactor matrix, let us just see what that was, that was minus 48, 40, minus 10, 18, minus 15, 4,

10, minus 8, 2 and you flip it over the diagonal, you take its transpose. So, your minus 48, 40, 10,

minus 10 became now the first column. So, just to check here, it is the first row,

here it is the first column and so on. So, your second row became the second column,

your third row became the third column, so 18, minus 15, 4 here is your second row that

becomes a second column and then 10, minus 8, 2 becomes a third column.

So, you have taken the transpose of the cofactor matrix. So, for your matrix A

we have computed the adjugate matrix. So, now let us do something funny, let us compute

A times 1 by determinant of A times adjugate of A. So, this 1 by determinant of A times adjugate of

A is supposed to be a matrix and 1 by determinant of A times adjugate of A

that matrix multiplied by A. So, let us do this computation,

well 1 by determinant of A times adjugate A, so we computed what is the determinant of A, that was 2, so that means you have to multiply adjugate by half, so if you multiply the adjugate by half the

first row becomes minus 24, 9, 5, the second row becomes 20, minus 15 by 2, minus 4, the third row

becomes minus 5, 2, 1 and we want to multiply on the left this same matrix by 2, by A.

So, your matrix A was 1, 2, 3, 0, 2, 8, 5, 6, 0 something you can check from your previous

slide. So, now let us multiply and see what we get, so let us let me do the one by one term and

then the rest I leave to you to check what we or some of the terms and the rest I leave to you to

check what we get. So, the first term is 1 times minus 24 plus 2 times 20 plus 3 times minus 15

that is minus 24 plus 40 minus 50, so that is 40 minus 39, that is a 1.

And then the next term, let us say the 1, 2 term, so 1 times 9 plus 2 times minus 15 plus 3 times 2

so that is 9 minus 15 plus 6, so 15 minus 15 that is a zero and you can carry on and do the other

terms so you are going to get 1, 0, 0, 0, 1 let us do this term just for fun. So, this is the 2, 3 term, so that is 0 times 5 plus 2 times minus 4 plus 8 times 1, clearly that is 0 and then the

rest is going to be 0, 0, 1. So, this is going to give you the identity matrix and now you can

do it the other way and you can check that you still get the identity matrix.

So, if you multiply minus 24, 9, 5, 20, minus 15 by 2, minus 4, minus 5, 2, 1 that is 1 by

determinant of A times adjugate A and multiply A on the right you are still going to get the

identity matrix. So, what does that tell you? That tells you that the inverse, A inverse is

that is for this example is 1 by determinant of A times adjugate of A,

well it turns out that this is a general fact and that is exactly what

we are going to state here. So, if A is an n by n matrix

and determinant of A is non-zero, then A inverse exists and equals 1 by determinant

of A times adjugate of A. So, how do we get this relationship? So, if you remember in the previous

video we had these 2 very nice formulae about cofactors. So, one was that summation aij cij, so this is a determinant of A where you take the sum from 1 through n. So, now let us divide both sides

by determinant of A, if you do that and then we will take the determinant of A inside the sum.

So, if you do that what we get is summation aij times 1 by determinant of A times

cij, j is 1 through n, this is 1 and now we can write this term over here as the ji-th term

of the adjugate, so this is the same as summation aij adjugate of A ji, why is that? Because cij

transpose, my bad, including the determinant of A, sorry, so this times 1 by determinant of A,

so that is because cij is the same as adjugate of A ji because adjugate of A

is the cofactor matrix transpose. So, this is 1 but this is exactly the,

so summation aij 1 by determinant of A adjugate of A

ji, so this is 1, so j is 1 through n, so what does this mean? This means that if you take the

ii-th entry of the matrix A times 1 by determinant of A

adjugate of A, if you take the ii-th, entry then the ii-th entry is 1, the diagonal entry is 1.

So now we can ask what happens to entries which are not on the diagonal. So, for that we will have to do summation aij c let us say kj, why do we care

about this, because if you work out in the same way, if you back calculate you will get here.

So, I will leave it to you to back calculate, so what I am saying is you take this matrix and look

at its ik-th entry, where i is not equal to k, then this is exactly what you are going to get,

well not this but this times determinant of A. So, this times 1 by determinant of A, but the point is

this thing over here is 0 and why is it 0, where i is not k, why is it 0? Well that is because we can

think of this as the determinant of a matrix in which the i-th row is the same as the i-th row of

A and the k-th row is also the same as the i-th row, so two rows are same in this.

So you can think of this as a determinant of a matrix with two rows the same, namely all rows of

A are exactly the rows of this matrix except the k-th one and the k-th one what do we have? We have

the i-th row. And in the i-th row what do we have? We have the i-th row. So, two rows are the same and we have seen in the last video that if two rows are the same then the determinant is 0,

so that is why this thing is 0. And from here it will follow that

the ik-th entry is also 0, so we will get that A times 1 by determinant A adjugate of A is

the identity matrix. Similarly, you can flip it around and work with columns and you will

get that is also the identity matrix, so that is how we get this formula for A inverse.

So, now let us get to what we really wanted to do, namely we want to find the solution of a linear,

system of linear equations where the coefficient matrix is invertible. So, I will repeat what

that means, the coefficient matrix is invertible means that it has an inverse which means the same

as its determinant is non-zero that is what we have seen, that if the determinant is non-zero,

then it has an inverse and of course, if it has an inverse the determinant is non-zero.

So, let us look at the system of linear equations Ax equals b where the coefficient matrix A is an

invertible matrix, so let us multiply both sides by A inverse, so if we multiply both sides by A inverse what do we get? So, here is Ax equals b, multiply both sides by A inverse

and when I multiply on the left notice that this is matrix multiplication, so it does not matter

whether I first look at A inverse A or first look at A times x.

So since it is to my advantage I will multiply A

inverse and A first and what do I get? I get the identity matrix,

so the identity matrix times x is A inverse b, but identity times anything is that thing itself,

so identity times x is x. So, we get x is A inverse b, so this is the solution. So, we wanted

to find the solution and here is the explicit form for the solution x is A inverse b.

So, let us do an example, so here is an example that we actually computed from much, much before,

so this was actually something to do with price of rice and dal and so on. So,

8x1 plus 8x2 plus 4x3 is 1960, 12x1 plus 5x2 plus 7x3 is 2215, 3x1 plus 2x2 plus 5x3 is 1135,

well I should say 1960, 2215 and 1135. So, let us form the corresponding system of linear equations, that is this is, this A is the coefficient matrix 8, 8, 4, 12, 5, 7, 3, 2, 5, x is the unknowns, the

column vector of unknowns and b is the column vector of the constants 1960, 2215

and 1135. So, now the first thing we want to check is what is determinant of A.

So, determinant of A we can compute this and the computation is, we know how to do this now it

gives you minus 188 which is non-zero, which is really all we care about. So once it is non-zero

that means A is invertible and we know explicitly what the solution is. It is A inverse times b,

so now what we have to do is we have to compute A inverse and then we have to multiply it with b.

So, how do we compute A inverse well? We have to find the minors,

this is what the minors are so 11, 39, 9, 32, 28, minus 8, 36, 8, minus 56 you will have to do this,

I trust now we can all compute determinants and so you will be able to do this. So,

just to reiterate to compute the 3, 2 minor you have to hide the third row and the second column and compute the determinant of the remaining matrix. So, from the minors we can get the

co-factor matrix, how do we do that? We multiply each of these by minus 1 to the power i plus j,

so that is how we get the cofactor matrix.

And then we can form the adjugate matrix, the adjugate is the transpose of the cofactor matrix,

so that is what we get and then A inverse is 1 by determinant of A times the adjective, so 1 by

minus 188 which is what we found the determinant to be times the adjugate matrix which is up here.

And then what is x? Which is the solution, the solution is A inverse b, so you multiply

all these and then finally what you get is 45, 125, 150 which means x1 is 45, x2 is 125, x3

is 150. Now, I will ask you to compare this with the solution you got, we worked this out when we

did systems of linear equations. So, look at that video and observe that we got the same thing.

So, finally we want to look at something called the homogeneous system of linear equations.

So, what is the homogeneous system of linear equations? When you write it as Ax equals b, so now A need not be a square matrix, so you could have m equations in n unknowns

but the b is 0, yeah, the constants appearing here, so these are all 0. So, this is 0,

all these are 0. So, if these are 0 it is called a homogeneous system of linear equations.

Now, if it so happens that you have a homogeneous system of linear equations,

so that means the matrix form is Ax is equal to 0 and this is a square matrix, so A square,

so I should have added that over here, so if A is an invertible matrix, then we know exactly

what the solution is, it is A inverse b but b is 0 here, so that means x is 0. So, a homogeneous

system of linear equations with n equations in n unknowns, so it has a unique solution 0,

if its coefficient matrix is invertible so that is the same as saying its determinant is non-zero and

here is something extra, it has an infinite number of solutions if its coefficient matrix is not

invertible, that means its determinant is 0. So, now we know explicitly that if you have n equations in n unknowns, then either and your system is homogeneous, so you are solving Ax is 0,

if the determinant of A is non-zero then the solution is A inverse 0 which is 0. And if the

determinant is 0, then you have an infinite number of solutions, so the first part is something we

did, the second part is something I am stating and we will do this in the next video. Thank you.